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Functional Meaning of Kernel-Selection Weights: Normalization, Branch Interventions, and Static Simplification in 3D Segmentation

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Abstract

Learned kernel-selection weights are often interpreted as indicating the importance of convolutional branches. Their functional reference, however, depends on the intervention and the operators surrounding the mixture. We analyze branch removal, renormalized removal, forced-path performance, and static simplification in a controlled three-dimensional segmentation system. A finite-epsilon identity characterizes when instance normalization attenuates removal-policy differences and when low surviving variance preserves them; a local derivative and counterexamples explain why weight magnitude alone cannot determine functional importance. Deterministic probes and direct measurements on fifteen terminal brain-tumor banks verify these operator conditions. Across five folds and three training seed groups, weight–removal associations exceed absolute weight–forced-path associations in every bank. Seed-specific pooled correlations are 0.879, 0.576, and 0.743 for removal, versus 0.094, −0.126, and 0.067 for forced paths. Paired removal policies differ little at the last decoder stage but by up to 0.281 region-macro Dice at the first; matched float32 sensitivity retains large differences. Separate five-fold, single-seed compact schedules reduce recorded active computation by approximately 53–58% relative to dense multi-branch controls, without resolving a uniquely preferred schedule. These observations support an operator-aware interpretation: selection weights can describe dependence within a jointly trained mixture without identifying a sufficient path or a uniquely preferred compact architecture.

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Keywords: adaptive kernel selection; instance normalization; functional importance; computational intervention; multi-branch networks; brain tumor segmentation

1. Introduction

Multi-scale convolutional systems use learned coefficients to combine branches with different receptive fields. In Selective Kernel Networks, channel-wise attention supports adaptive receptive fields [1]. These explicit, inexpensive-to-record coefficients refer to named components, making them attractive diagnostics. Interpreting them requires a specified functional target.

Byra et al. provide a direct medical-imaging precedent: in breast-ultrasound selective-kernel U-Net, sample-level dilated-branch attention correlated with lesion size (Spearman

coefficients around 0.7 in the early expansion-path SK blocks) [2]. This association describes a learned morphological response. We ask which functional target it reflects: removal dependence, forced-path performance, or separately trained compact-model performance.

A large coefficient might identify a branch whose removal changes the system, one that works well alone, or one preferable in a compact architecture. Removal preserves the remaining fitted parameters; forced-path inference retains one branch and the shared encoder/downstream network; compact-model training changes the optimization problem. A *static kernel schedule* fixes one kernel per selected stage before training. Unlike fitting constant output-mixture weights over a retained expert bank, it changes structure. These operations require separate outcome definitions and comparisons.

Attention-interpretation studies motivate this distinction. Jain and Wallace compare attention with functional attribution and construct alternative distributions with similar predictions [3]. Serrano and Smith manipulate attention in trained predictors to relate its magnitude to prediction changes [4]. Wiegreffe and Pinter emphasize explanation targets and model-consistent controls [5]. These token-attention studies motivate precise kernel-mixture definitions without determining medical-convolutional behavior.

The mixture is rarely the final prediction: normalization and activation mediate downstream differences. Instance normalization centers and rescales spatial variation independently per input and channel [6], potentially making distinct weight edits nearly equivalent. Conversely, a small weight can multiply a large feature tensor or a strongly amplified direction. Coefficients must therefore be examined with their operators.

We make three connected contributions: define removal dependence, forced-path performance and independently trained kernel competence, deriving a finite-epsilon identity and local sensitivity expression; test their operator conditions with deterministic tensors and the unchanged selector; and verify the analysis through terminal-checkpoint brain-tumor interventions and static architectural simplification. The object is a kernel-selection signal within a specified computational system.

2. Related Work

2.1. Adaptive receptive fields and kernel mixtures

CondConv and Dynamic Convolution assemble input-conditioned kernel parameters before convolution [7,8], unlike executing all branches of a feature-fusion bank. CondConv uses sigmoid routing and Dynamic Convolution simplex weights; our derivations require the stated simplex assumptions.

Dynamic Convolution replaces trained attention aggregation with averaging, maximum-weight selection or shuffled weights and measures classification accuracy [8]. Our candidate-wise stage edits instead measure segmentation Dice and compare removal with renormalized removal under shared post-fusion instance normalization; compact-model training is evaluated separately.

SK units combine branch features with channel-wise softmax weights [1]. We implement this selection concept in a three-dimensional U-Net [9] with four branches and shared post-fusion instance normalization. Different normalization placement, branches and task make this a controlled characterization, not a reproduction of the original SKNet mechanisms or performance.

Medical applications include Byra et al.'s SK-U-Net, which combines conventional and dilated convolutions and examines attention–mass-size associations [2], and ESKNet, which adds spatial and channel recalibration, residual learning, and deep supervision for breast ultrasound [10]. More recently, SGDC uses an explicitly supervised structure branch to condition spatially varying dynamic kernels and gating [11]. These medical applications use different operators: spatially varying coefficients and residual paths fall outside our

simple post-fusion per-channel equivalence. Their performance results establish application relevance, not branch-removal evidence.

2.2. Attention and functional attribution

The debate on attention explanation includes correlation tests, removal experiments, alternative weights, seed calibration, and jointly trained adversaries [3–5]. These controls address different targets. We examine intervention–normalization interactions in kernel mixtures and distinguish frozen-system dependence from compact-model training.

Serrano and Smith zero selected token-attention weights, renormalize the survivors, and measure output-distribution divergence and decision flips [4]. Our paired-policy analysis asks how subsequent per-channel spatial normalization attenuates or preserves the difference between removal with and without renormalization. It adds an operator condition and trained-stage measurements to the established erasure test.

Michel et al. mask or retain single attention heads and study incremental pruning [12], directly preceding the removal/forced-path distinction. We examine normalized spatial mixtures with paired removal policies and repeated training. Adebayo et al. use parameter and label randomization to test whether saliency maps depend on the learned system [13], a complementary property to removal loss or forced-path performance.

ROAR retrains image classifiers after removing ranked input pixels to reduce confounding from the modified input distribution [14]. Our schedules instead change internal structure while retaining imaging inputs. Both distinguish frozen-system edits from training a modified system.

2.3. Normalization and representational equivalence

Instance normalization removes instance-specific contrast in image generation [6]. From its standard definition, we derive and verify the finite-epsilon relation governing paired branch removal and its equivalence conditions; no new normalization operator is introduced.

Scale invariance and optimization are also linked in van Laarhoven's analysis of weight rescaling under batch, weight, and layer normalization [15], linking parameter scale to effective learning rate. Here, branch removal rescales a fixed-checkpoint mixture under per-channel spatial normalization with input-specific evaluation statistics.

Norm-based Transformer analyses already show that attention coefficients must be considered together with transformed-vector magnitude [16]. A subsequent study incorporates residual connections and layer normalization into the entire attention block [17]. This established operator-aware attribution motivates our narrower object: spatial per-channel kernel mixtures, paired removal-policy equivalence, forced paths and separately trained static architectures.

3. Mathematical Formulation

3.1. Mixture and intervention targets

For input x at stage s, branch k produces a tensor $b_{s,k}(x) \in \mathbb{R}^{C_s \times V_s}$. The number of spatial positions is V_s. Channel-wise weights satisfy

$$a_{s,k,c}(x) \geq 0, \qquad \sum_{k=1}^{K} a_{s,k,c}(x) = 1, \qquad z_{s,c}(x) = \sum_{k} a_{s,k,c}(x) b_{s,k,c}(x). \tag{1}$$

Each coefficient is constant across spatial positions within its channel. The stage output is $u_s = \phi_s(N_s(z_s))$. Let F denote the full trained network, including all unchanged stages, and $\ell(F(x), y)$ its task loss. Weights are generated before an edit at the chosen stage and retained for the surviving branches. Later stages may respond to the modified representation.

For a chosen branch k, define the unnormalized removal mixture and the renormalized removal mixture by

$$z_{s,c}^{-k} = \sum_{j \neq k} a_{s,j,c} b_{s,j,c}, \qquad \widetilde{z}_{s,c}^{-k} = \frac{z_{s,c}^{-k}}{1 - a_{s,k,c}}. \tag{2}$$

The latter is defined when $a_{s,k,c} < 1$. A forced-path intervention instead uses $z_{s,c}^{[k]} = b_{s,k,c}$. For loss-based reporting, removal dependence is

$$D_{s,k}(x, y) = \ell(F_{s,-k}(x), y) - \ell(F(x), y), \tag{3}$$

while forced-path performance is $Q_{s,k}(x, y) = M(F_{s,[k]}(x), y)$ for a specified higher-is-better metric M. Dice-based removal effects use $M(F(x), y) - M(F_{s,-k}(x), y)$. A negative effect means that removal improved this metric; the definition imposes no positivity assumption.

An independently trained fixed-kernel model H_k has competence $M(H_k(x), y)$. This is distinct from $Q_{s,k}$: $F_{s,[k]}$ retains a co-adapted shared network and changes only one stage. We reserve the term forced-path performance for the latter throughout.

3.2. Exact relation under instance normalization

For a vector $z \in \mathbb{R}^V$, let $\mu(z)$ and $v(z)$ be its spatial mean and population variance. The non-affine normalization core is

$$N_\epsilon(z) = \frac{z - \mu(z)\mathbf{1}}{\sqrt{v(z) + \epsilon}}. \tag{4}$$

The following relation concerns one input and channel; its scale may differ between channels or inputs.

Proposition 1 (Finite-epsilon scaling relation)**.** *For $\epsilon \geq 0$, $t > 0$, and $v(z) + \epsilon > 0$,*

$$N_\epsilon(tz) = r(t, v, \epsilon) N_\epsilon(z), \qquad r(t, v, \epsilon) = \sqrt{\frac{v + \epsilon}{v + \epsilon/t^2}}. \tag{5}$$

When $\epsilon = 0$ and $v > 0$, the two normalized tensors are identical. For $t \geq 1$ and $v > 0$,

$$0 \leq r - 1 \leq \frac{\epsilon(1 - t^{-2})}{2(v + \epsilon/t^2)}. \tag{6}$$

Proof. The mean of tz is $t\mu(z)$ and its variance is $t^2 v(z)$. Substitution into N_ϵ gives Equation (5). For the bound, write $r = \sqrt{1 + q}$ with $q = \epsilon(1 - t^{-2})/(v + \epsilon/t^2) \geq 0$, and use $\sqrt{1 + q} - 1 \leq q/2$. □

At $v = 0$ and $\epsilon > 0$, $r = t$ but both normalized tensors are zero. Thus a large scaling ratio alone does not imply a large absolute tensor difference; the bound in Equation (6) can also be loose at low positive variance.

Equation (2) supplies $t_c = (1 - a_{s,k,c})^{-1} \geq 1$. Consequently, removal and renormalized removal have identical post-normalized stage outputs in the zero-epsilon, positive-variance idealization. Applying the same activation and downstream computation preserves this equality. With finite epsilon, the ratio of spatial variance to epsilon determines the remaining amplitude difference. The statement concerns the stage tensor: a finite tensor difference does not guarantee an identical thresholded mask or equal Dice. Downstream sensitivity and decision margins still matter.

For an affine normalization $A_\epsilon(z) = \gamma N_\epsilon(z) + \beta$, the corresponding difference is $\gamma(r - 1)N_\epsilon(z)$; the entire affine output is not generally multiplied by r. The implemented shared post-fusion normalization is non-affine and uses input-specific statistics during

evaluation. The executable removal routine floors a positive surviving-weight denominator at 10^{-8} and returns a zero mixture if no branch survives. In the floored case, Equation (5) uses the actual positive scale from that routine; the unfloored expression $t = (1 - a_k)^{-1}$ applies above the floor. Spatially varying weights, normalization across multiple channels with different t_c, or fixed running statistics fall outside the stated per-channel equivalence conditions.

3.3. Boundary under group normalization

Group normalization pools spatial positions and channels within each predefined channel group [18]. For one input and group g, flatten its features as $z_g \in \mathbb{R}^{m_g}$ and let $P_g = I - \mathbf{1}\mathbf{1}^\top/m_g$ center that group. Renormalized removal applies a positive diagonal scale D_g whose entries repeat the channel-specific t_c across spatial positions. Where the denominator is positive, its non-affine normalized core is

$$\mathrm{GN}_{g,\epsilon}(D_g z_g) = \frac{P_g D_g z_g}{\sqrt{\|P_g D_g z_g\|^2/m_g + \epsilon}}. \tag{7}$$

If every channel in the group has the same scale $t > 0$, then $D_g = tI$; substitution gives exactly Equation (5) with the group variance in place of the per-channel spatial variance. For unequal scales, however, $P_g D_g \neq D_g P_g$ and the group mean is $\mu_g(D_g z_g) = m_g^{-1} \sum_i (D_g)_{ii} (z_g)_i$. A common factor cannot generally be extracted from the centered tensor or its variance. This is the step at which the instance-normalization argument ceases to apply; even at zero epsilon, the two outputs need not be proportional. Special inputs can still yield equality.

One channel per group recovers the instance-normalization core. The all-channel, one-group control in Figure 1B instead combines channels with unequal surviving-mass scales and illustrates this boundary. The distinction is therefore the alignment of rescaling with the statistics pool, rather than the normalization name alone. A subsequent fixed affine transform is handled as above; the common-scale relation is a statement about the normalized core.

3.4. Local functional sensitivity

Let $C = I - \mathbf{1}\mathbf{1}^\top/V$, $w = Cz$, and $q = \sqrt{\|w\|^2/V + \epsilon}$. On the domain $q > 0$, differentiation yields

$$DN_\epsilon(z)[h] = \frac{Ch}{q} - \frac{w\langle w, Ch\rangle}{Vq^3}. \tag{8}$$

In particular,

$$DN_\epsilon(z)[z] = \frac{\epsilon Cz}{(v+\epsilon)^{3/2}}. \tag{9}$$

For a group-normalized core, Equation (8) has the same form after replacing C, V by P_g, m_g. A relative channel rescaling gives $h = Az_g$, with diagonal A; this direction need not lie in that derivative's null space even at zero epsilon. For instance normalization, at zero epsilon and positive variance, infinitesimal common scaling lies in the normalization derivative's null direction. The equation describes the geometry of the operator; it does not imply that removing a branch is merely common scaling, since removal also changes the mixture's direction.

For a weight transfer from branch j to branch k, hold the selected-stage branch tensors fixed and choose δa so that the coefficients remain in the simplex; at its boundary only admissible directions are considered. Then $\delta z = \delta a(b_k - b_j)$. Assume $q > 0$ and that both

ϕ and the downstream loss are differentiable at the evaluated representation, excluding activation kinks. The first-order change is

$$\delta\ell = \delta a \left\langle \nabla_u \ell, D\phi(N_\epsilon(z)) D N_\epsilon(z)[b_k - b_j] \right\rangle + o(|\delta a|). \tag{10}$$

The downstream gradient includes the response of subsequent stages; their selectors are not frozen by the selected-stage edit. Weight magnitude does not specify the branch difference, normalization Jacobian, or downstream loss gradient. A full removal or forced-path edit is a finite intervention and need not be approximated well by this derivative. For thresholded Dice, which is not differentiable in the required sense, Equation (10) motivates the distinction but is not used to calculate the reported effects.

3.5. Coefficient magnitude does not order forced-path performance

Example 1 (Cancellation with positive simplex weights)**.** *Consider an identity readout, squared loss, target $y = 0$, coefficients $(0.6, 0.3, 0.1)$, and scalar branch values $(5, 0, -30)$. The mixture equals zero and has zero loss. Forced-path losses are $(25, 0, 900)$, so the largest-weight branch is not the best forced path. Unnormalized removal losses are $(9, 0, 9)$, so the smallest-weight branch has the same removal effect as the largest-weight branch.*

This construction excludes a universal coefficient ordering of either target, without estimating cancellation prevalence in medical networks. Interpretation requires branch features and a downstream target. Segmentation branches are spatial features, not calibrated tumor probabilities: their mixture passes through normalization, activation and a shared decoder. The scalar identity readout isolates cancellation without spatial normalization or anatomical kernel assignments. We therefore assess removal and forced-path Dice within the actual normalized network, and independently train fixed schedules to assess compact-model performance. Shared branch names do not equate these three operations.

Example 2 (Cancellation in normalized spatial features)**.** *Consider two intermediate feature maps,*

$$u = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

Each has zero mean and variance $1/2$, and their inner product is zero. Use branch maps $(5u, v, -30u)$ and weights $(0.6, 0.3, 0.1)$. Their mixture is $0.3v$. Apply the non-affine instance-normalization core with zero epsilon, followed by an identity readout, and set the feature target to $y = N_0(v)$. With loss $\ell(z) = \|N_0(z) - y\|^2/4$, the mixture loss is zero and forced-path losses are $(2, 0, 2)$. All normalized maps in this comparison have positive variance. The largest-weight branch is therefore not the best forced path even after spatial normalization.

Example 2 extends the scalar construction to spatial normalization. Its normalized-feature target and squared-error loss are neither MRI observations nor Dice, and assign no anatomical kernel role.

4. Materials and Methods

4.1. Numerical verification

We implemented Equation (5) with float64 tensors and a fixed random seed of 20261001. The variance-to-epsilon ratio spans 10^{-6} to 10^{8} at 71 logarithmically spaced points; scales are 1.1, 2, and 8, with $\epsilon = 10^{-5}$. Centered tensors are rescaled to the specified population variance before each comparison. The residual is the maximum absolute difference between direct normalization and the analytic prediction.

A second experiment fixes branch tensors and weight edits while changing only the surrounding operator: post-fusion instance normalization with epsilon zero or 10^{-5}, no post-fusion normalization, branch-local normalization only, group normalization across all channels, and spatially varying coefficients followed by instance normalization. Twelve seeded tensor trials use two instances, four branches, four channels, and a $5 \times 6 \times 7$ spatial grid. We report the relative tensor norm change between removal and renormalized removal.

We also evaluate the unchanged SK selector on twelve small CPU inputs, using deterministic random head weights rather than uniform initialization. Replacing activation by identity only in this probe verifies the pre-activation finite-epsilon relation. These numerical controls leave trained models and checkpoints unchanged.

4.2. Brain-tumor cohort and common training

Medical-imaging experiments use the BraTS 2023 adult glioma training cohort [19,20]. Our data manifest contains 1,251 studies from 1,133 patients, including 118 patients with two studies. A fixed five-fold partition groups studies by patient. The primary stage study evaluates all five folds for three base seeds (S1: 2026; S2: 20261001; S3: 20261011), using base seed plus fold index: fifteen terminal banks on the same 1,251 studies, not independent cohorts. The historical schedule study used folds 0–2 (751 studies, 679 patients); a retrospective missing-fold extension completes five-fold schedule and final-block pooling comparisons, each still single-base-seed. The separate single-configuration architecture matrix also covers five folds.

S1 and schedule-study SK share base seed 2026 and recorded scientific settings, but use separately trained checkpoints: new folds 0–4 versus historical folds 0–2 plus the later extension. Appendix D compares their scores and provenance; nominal seed/recipe agreement does not establish bitwise training equivalence.

A separate routing manuscript in preparation shares this cohort, preprocessing, patient folds and historical fixed-kernel infrastructure. It addresses complete-expert competence, case-level role reproducibility and adaptive/static output policies; this paper addresses kernel interventions, normalization and static schedules. The fifteen new banks are not routing-paper replicas, nor are shared patients/models independent cross-paper validation. Each paper establishes its primary findings independently.

Four MRI modalities are cropped using the union of their nonzero anatomy plus a four-voxel margin and padded to $192 \times 224 \times 192$ without truncating the anatomical crop. Labels do not set crop bounds. Each modality is standardized within its nonzero support and clipped to $[-5, 5]$. Atomic labels are converted to whole tumor (WT), tumor core (TC), and enhancing tumor (ET). The common recipe uses full volumes, batch size four, the GPU augmentation specified in Appendix B, binary cross-entropy plus batch Dice, AdamW with learning rate 2×10^{-4} and weight decay 10^{-5}, cosine decay, 100 epochs, and bfloat16 training. Terminal checkpoints provide the primary evaluation. The evidence manifest compares versioned and actual configurations; all twenty-five schedule model–fold records agree on non-path fields. Appendix D links recipes, terminal evaluations and FLOPs.

Table 1 specifies the common backbone and selector locations. All backbone and post-fusion InstanceNorm3d layers use $\epsilon = 10^{-5}$, `affine=False`, and `track_running_stats=False`: evaluation uses the current input's spatial statistics, not accumulated training statistics. LeakyReLU has negative slope 0.01. The selector replaces the second convolution, normalization, and activation of an indicated decoder block, leaving its first convolution and skip connection unchanged.

Table 1. Common U-Net architecture and full-volume selector resolutions. Spatial dimensions refer to the preprocessed input, not random patches.

Component	Implementation
Input and output	Four MRI channels; three output logits (WT/TC/ET).
Encoder	Five levels with channels 32, 64, 128, 256, 512; two bias-free 3^3 convolutions per level, each followed by InstanceNorm and LeakyReLU.
Down/up sampling	Four stride-two max pools; four 2^3, stride-two transposed convolutions and concatenated encoder skips.
Decoder and head	Channels 256, 128, 64, 32; two convolution–norm–activation blocks before replacement; final 1^3 convolution.
Selector positions	Decoder 1: $128 \times 48 \times 56 \times 48$; decoder 2: $64 \times 96 \times 112 \times 96$; decoder 3: $32 \times 192 \times 224 \times 192$ (channels $\times$ spatial shape).
SK selector	Sum branch features, global average pooling, hidden width $\max(8, C/4)$, ReLU, four channel-wise heads and branch softmax. Avg+max concatenates average and maximum pooled features.

4.3. Branches, stages, and functional measurements

The branch bank contains dense 3^3, 5^3, dilation-two 3^3, and $1 \times 3 \times 3$ anisotropic convolutions. Branches have no local normalization; shared instance normalization and a leaky rectifier follow fusion. Multi-stage SK inserts selectors at decoder indices 1, 2, and 3. SK produces channel-wise weights from pooled fused branch features; the separately reported final-block dense-soft variant produces scalar weights from pooled shared input.

At each selected stage, every branch is removed, removed with renormalization, or forced alone on the same terminal checkpoint and validation studies; other stages compute normally. Weights are averaged over channels, then studies. The positive-reference region-macro endpoint averages study Dice over present references within each region, then equally over three regions. Spearman correlations use twelve stage–branch cells per fold; Appendix C disaggregates four-branch within-stage correlations and functional/weight ranges using the same cells and endpoint. Retrospective stage sensitivities residualize within-bank ranks on stage indicators, rank branches independently within each stage, and omit each stage in turn. We compare signed weight–removal association with absolute weight–forced-path association (definitions/exceptions in Appendix C). Taking the latter's absolute value counts a negative association as a competing signal of equal magnitude, although it opposes the interpretation that larger weights identify better forced paths. This sets a hurdle at least as high as a signed-to-signed comparison for the positive removal association. These and the 60-cell seed-specific pooled correlations are descriptive: folds share training data and stages a fitted network, precluding independent-replica or patient-selection interpretations. Cross-seed winner agreement compares maximum mean weight, removal effect or forced-path Dice in 15 matched fold–stage pairs, resolving ties by fixed branch order. This measures bank-level, not patient-level, ranking stability.

4.4. Direct checkpoint-normalization measurements

Observation-only hooks capture selected-stage pre-normalization mixtures and post-normalization/pre-activation outputs for paired removals on all fifteen banks. Full-cohort evaluation preserves BF16 autocast, batch size four and batch membership. Fidelity checks show identical observed/unobserved repeats and historical Dice; paired stage inputs, branch tensors and pre-edit weights must match. Measurements comprise surviving mass, denominator-floor events, variance/epsilon and residuals of $\widetilde{z} = tz$ and Equation (5). Fixed captured features are replayed in float64 at epsilon zero, 10^{-8}, 10^{-5} and 10^{-3}, without checkpoint edits.

A matched sensitivity selects, before inference, the first two patients per fold ordered by SHA-256 of a protocol-specific prefix and patient ID. All their studies give ten pa-

tients/eleven studies, reused across three seeds. Fresh BF16 and full FP32 runs use batch size one with TF32 disabled. They change upstream representations as well as normalization arithmetic; only fixed-feature local replay isolates epsilon. This post-result audit does not enlarge the patient sample through channels or repeated seeds.

4.5. Static kernel schedules and uncertainty

Two compact architectures were derived during the earlier 30-epoch investigation: an attention-derived anisotropic/anisotropic/3^3 schedule and a drop-derived dilated-$3^3/3^3/3^3$ schedule at decoder indices 1/2/3. Each was subsequently trained for 100 epochs on folds 0–2 and compared with multi-stage SK, multi-stage uniform fusion, and canonical U-Net under the same nominal recipe. The later thirteen-fit extension adds folds 3–4 to both schedules, multi-stage SK, multi-stage uniform and GAP dense-soft, and folds 2–4 to average-plus-maximum dense-soft. The completed schedule comparison uses five models on five folds, base seed 2026 and effective seed $2026 + f$. This completes fold coverage, not three-seed replication. Schedule construction used the original folds and extension followed result inspection, making it a discovery/internal-repetition study.

We recalculate paired intervals from terminal study-level outputs using 4,000 patient-cluster bootstrap draws within each fold, with seed 20261001. Both studies from a sampled repeated-study patient receive the same multiplicity. Each draw recomputes positive-reference region means and then the equal-fold mean. Explicitly, let I_{fir} indicate a positive reference for study i, region r, and fold f, and let d_{mfir} be its Dice for model m. If $w_{fi}^{(b)}$ is the bootstrap multiplicity of study i's patient, the draw-level score and paired contrast are

$$S_m^{(b)} = \frac{1}{3F} \sum_{f=1}^{F} \sum_{r \in \{\mathrm{WT,TC,ET}\}} \frac{\sum_i w_{fi}^{(b)} I_{fir} d_{mfir}}{\sum_i w_{fi}^{(b)} I_{fir}}, \qquad \Delta_{mn}^{(b)} = S_m^{(b)} - S_n^{(b)}, \quad F = 5. \tag{11}$$

Models share sampled patients and multiplicities; paired-contrast percentiles 2.5 and 97.5 define the 95% interval. Study IDs/reference-presence flags match across all five models, with no missing predictions; zero eligible denominators raise an error. Unit multiplicities give the point estimate. Equal fold and WT/TC/ET weighting yield eligible-region study means, not a pooled patient-weighted score; Appendix D lists study/patient denominators. Intervals condition on fitted models, neither quantifying retraining variability nor proving equivalence. Active FLOPs describe the full-volume profile, not latency. The repository profiler uses THOP 0.1.1 (environment lock build `0.1.1.post2209072238`), in evaluation mode on one float32 input of shape $1 \times 4 \times 192 \times 224 \times 192$, with FLOPs $= 2 \times$ MACs. Dense SK and uniform fusion execute every branch, so their active and total counts coincide; a compact fixed schedule constructs and executes only its chosen branch at each selector. Counts follow THOP's operator-accounting convention, not measured hardware instructions or training cost. Recorded counts differ between the historical and later profiling regimes by a common 0.443925 TFLOPs across these five families despite the same reported shape and counting rule. We retain the observed ranges and compute reductions against the same-fold dense comparator, rather than presenting these records as harmonized re-profiling. The offset exactly matches output-grid versus input-grid operation counting for the four stride-two transposed convolutions; this arithmetic match does not establish which profiler version was imported in each historical run.

During the preparation of this work, the authors used OpenAI Codex and ChatGPT for software development, analysis support, and manuscript preparation. The authors reviewed and edited the outputs and take full responsibility for the content of this publication.

5. Results

5.1. Numerical identity and architectural boundaries

Maximum absolute finite-epsilon residuals are `1.33e-15` on the float64 variance/scale grid and `2.66e-15` in the executable selector. Figure 1 shows accurate approximate equivalence at ordinary variance and failure near epsilon. Zero-epsilon per-channel post-fusion normalization cancels common positive rescaling; removing it, moving it to branches, pooling differently scaled channels or using spatial weights generally does not.

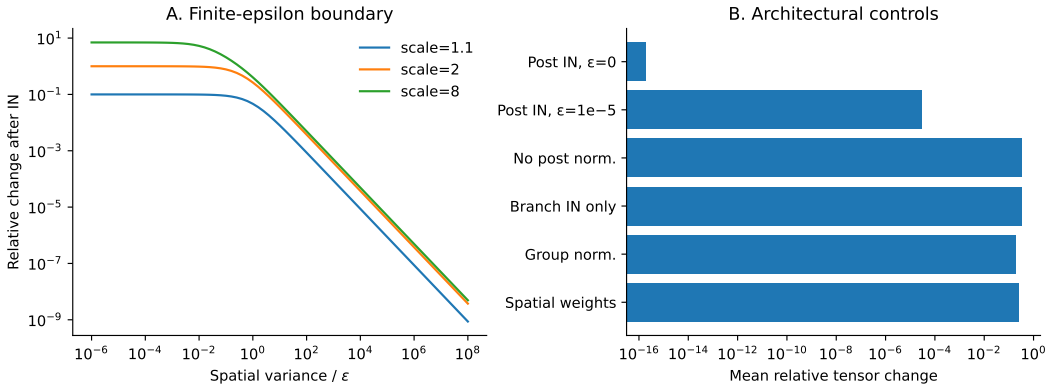


Figure 1. Numerical verification of removal-policy equivalence. **A**: Relative change after instance normalization versus spatial variance divided by epsilon, for three positive scales. **B**: Mean relative tensor change across twelve trials under controlled operator changes. The group-normalization control pools all four channels in one group; Section 3.3 explains why unequal channel scales prevent the per-channel argument from applying. Zero-epsilon post-fusion instance normalization approaches floating-point precision; the other conditions identify boundaries of the per-channel constant-scale argument.

5.2. Trained checkpoints expose the finite-epsilon boundary

The full-cohort audit reconstructs all 180 stage–branch endpoints and their paired channel measurements without changing the original predictions. Table 2 shows strong stage dependence. Across sixty cells per stage, median absolute removal-policy differences are 1.58×10^{-4}, 1.14×10^{-6} and 4.40×10^{-7} Dice at decoder stages 1, 2 and 3, respectively. The corresponding maxima are 0.280635, 0.000376 and 0.000019. Thus, small typical differences do not justify treating the two policies as equivalent throughout the network. At stage 1, absolute differences exceed 0.001, 0.01 and 0.05 in 17/60 (28.3%), 6/60 (10.0%) and 3/60 (5.0%) cells, respectively; no stage-2 or stage-3 cell exceeds 0.001. The three differences above 0.05 occur in separate seeds and folds, all for the dilated branch (Appendix E). These are descriptive experiment-cell frequencies, not patient-level event rates. The largest full-cohort difference occurs for removal of the dilated branch at stage 1 in S1/fold 0: unnormalized removal gives 0.816314 Dice and renormalized removal 0.535680. The matched batch-one precision subset retains a stage-1 maximum difference of approximately 0.40 under both BF16 and FP32, contradicting a solely BF16 explanation.

Table 2. Direct checkpoint comparison of removal policies on fifteen banks. Δ is renormalized-removal minus unnormalized-removal positive-reference region-macro Dice; each entry summarizes sixty seed–fold–branch cells at one stage. Primary results use the full cohort and original batch-four BF16 runtime. Subset columns use the same ten patients/eleven studies in batch-one BF16 and FP32, not a full-cohort FP32 test. The lower panel reports full-cohort strict exceedances, $n/60$ (percentage), at three retrospectively chosen descriptive thresholds.

Decoder	Primary median $\lvert\Delta\rvert$	Primary max $\lvert\Delta\rvert$	Subset BF16 max	Subset FP32 max
1	1.58e-04	0.280635	0.400841	0.402439
2	1.14e-06	0.000376	0.000301	0.000364
3	4.40e-07	0.000019	0.000103	0.000082

Decoder	$\lvert\Delta\rvert > 0.001$	$\lvert\Delta\rvert > 0.01$	$\lvert\Delta\rvert > 0.05$
1	17/60 (28.3%)	6/60 (10.0%)	3/60 (5.0%)
2	0/60 (0.0%)	0/60 (0.0%)	0/60 (0.0%)
3	0/60 (0.0%)	0/60 (0.0%)	0/60 (0.0%)

At stage 1, low surviving variance/epsilon permits near-unit post-normalization RMS differences despite nearly proportional input mixtures. Finite-epsilon predictions closely match the outputs, and fixed-feature zero-epsilon replay largely removes the difference. Appendix E reports channel counts, variance ranges, floor events and residuals. This verifies the captured-stage mechanism, not individual-channel attribution of final Dice or every downstream amplification mechanism.

5.3. Weights track removal dependence more than forced paths

Table 3 reconstructs all fifteen banks' terminal correlations. Weight–removal correlations are positive in every bank ($\rho = 0.252$–0.951) and exceed the absolute weight–forced-path correlation in every bank ($|\rho| \leq 0.273$). The descriptive pooled removal associations are 0.879, 0.576, and 0.743 for S1, S2, and S3, respectively; forced-path associations are 0.094, -0.126, and 0.067: the distinction repeats with seed-dependent strength. Figure 2 displays the cells; global correlations do not imply exact study-specific rankings. Similar paired removal correlations also do not imply tensor/mask equivalence, given the stage-1 exceptions in Section 5.2. Forced paths change mixture content and test a different target.

Within-stage removal association is positive in 41 of 45 fold–stage cells (range -0.6 to 1.0), while forced-path associations have both signs (Appendix C). These descriptive four-branch correlations show heterogeneous stage-specific ranking strengths. Removal-effect ranges also differ: similar correlations do not imply equal effects.

After residualizing ranks on stage indicators, removal associations remain positive and exceed absolute forced-path associations in all fifteen banks ($\rho_{\text{removal}} = 0.230$–$0.966$). Independently ranking the four branches within each stage retains the contrast in 14/15 banks. Omitting stages 1, 2, or 3 retains it in 13/15, 15/15, and 14/15 banks, respectively; removal association stays positive throughout (Table A3). The overall distinction thus persists without uniform within-stage rankings.

Equal-fold mean Dice is 0.904036, 0.902962, and 0.902774 for S1–S3. Across the 15 matched fold–stage groups, all three seeds identify the same highest-weight branch in 9/15 groups, the same largest-removal-effect branch in 11/15, and the same best forced path in 6/15. Pairwise agreement counts for S1/S2, S1/S3, and S2/S3 are respectively 9/15, 13/15, and 11/15 for weight; 12/15, 13/15, and 11/15 for removal; and 12/15, 7/15, and 7/15 for forced paths. These counts show partial, target-dependent ranking reproducibility, not a universally stable or unstable named-kernel preference.

Table 3. Spearman association between stage–branch mean weights and functional measurements on terminal multi-stage SK checkpoints. There are twelve cells per bank and 60 cells in each seed-specific pooled row. Rows S1/0–4, S2/0–4, and S3/0–4 identify base seeds 2026, 20261001, and 20261011 and folds 0–4. Pooled rows are descriptive summaries, not independent replicates.

Fold	Drop	Renormalized drop	Keep-one
S1/0	0.951	0.937	0.021
S1/1	0.783	0.594	0.105
S1/2	0.944	0.902	-0.028
S1/3	0.909	0.909	0.021
S1/4	0.734	0.706	0.042
S2/0	0.252	0.252	-0.203
S2/1	0.552	0.573	-0.238
S2/2	0.790	0.762	-0.147
S2/3	0.587	0.559	-0.224
S2/4	0.804	0.797	-0.007
S3/0	0.503	0.503	-0.021
S3/1	0.664	0.664	0.007
S3/2	0.804	0.741	0.273
S3/3	0.944	0.944	-0.119
S3/4	0.734	0.734	-0.077
S1 pooled	0.879	0.842	0.094
S2 pooled	0.576	0.562	-0.126
S3 pooled	0.743	0.755	0.067

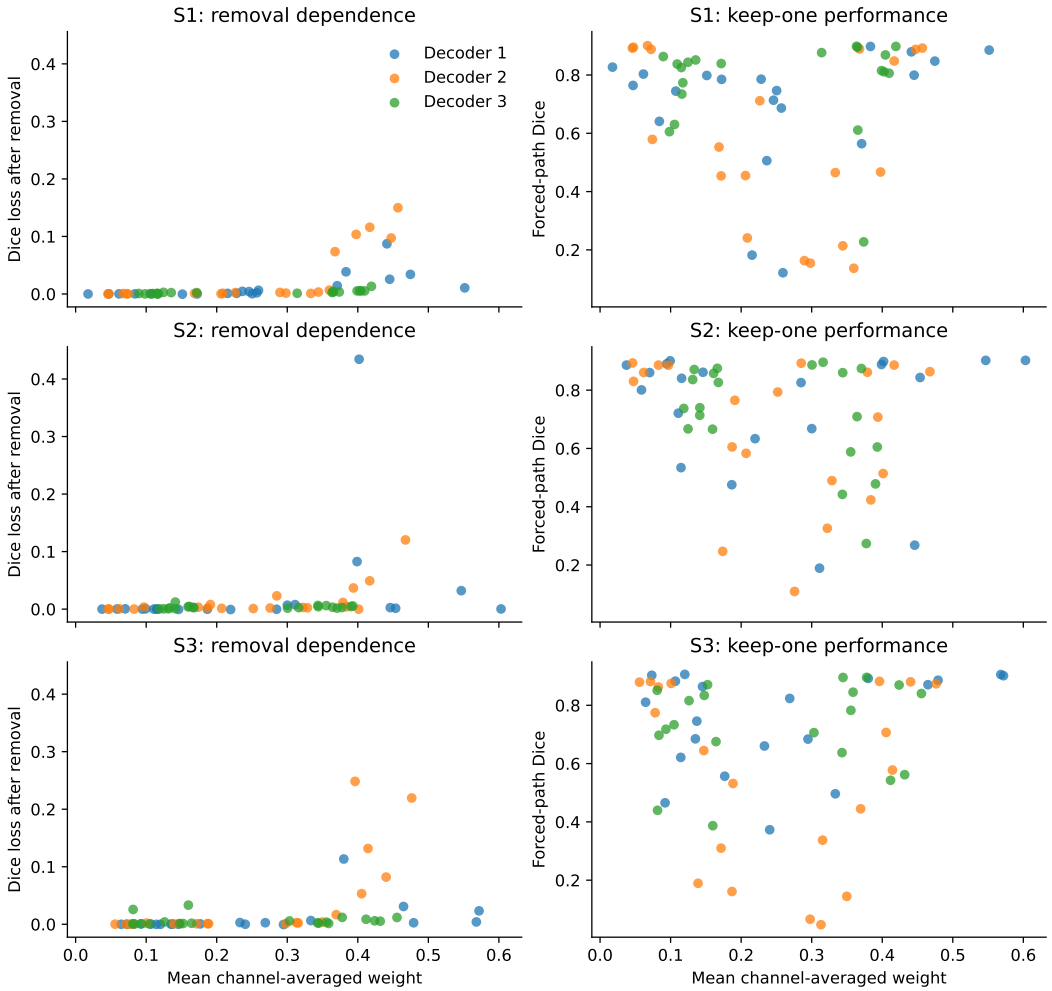


Figure 2. Mean weight and two functional references on the same SK banks. Rows separate the three base-seed groups. Each point is one fold–stage–branch cell; colors distinguish decoder stages. Removal dependence and forced-path Dice exhibit different relations with the recorded coefficients. The points are grouped measurements, not independent patients or retraining replicas.

In the historical three-fold study, the earlier and terminal training horizons also favor different named stage schedules. The 30-epoch attention-derived schedule is anisotropic/anisotropic/3^3, whereas terminal mean attention favors dilated-3^3/dilated-3^3/anisotropic. Terminal mean forced-path Dice favors dilated-3^3/5^3/5^3. These are distinct summaries of a changing trained system rather than a single identified kernel assignment.

5.4. Static simplification and its comparator

The completed five-fold schedule comparison is reported in Table 4. Equal-fold Dice is 0.903489 for the attention-derived schedule, 0.902562 for the drop-derived schedule, 0.902913 for multi-stage SK, 0.902626 for uniform fusion and 0.902464 for canonical U-Net. The SK value here belongs to the separate schedule-study bank, not the S1 replication bank (0.904036); Appendix D gives the run-level correspondence. Relative to the dense multi-branch profiles of 8.56–9.00 TFLOPs per volume, attention-derived and drop-derived schedules record 3.57–4.01 and 3.79–4.24 TFLOPs, respectively. Same-fold recorded reductions are 55.5–58.3% and 52.9–55.7%. Canonical U-Net records 3.79–4.24 TFLOPs, so the savings are chiefly relative to retaining multiple dense branches at each selected stage. The result does not establish that the diagnostic derivation found a uniquely superior compact model.

Table 4. Completed five-fold, 100-epoch, single-base-seed static schedule study. Dice is the equal-fold mean of study-wise positive-reference region-macro Dice. Parameters and active FLOPs are read from the associated run artifacts. FLOP ranges retain the historical/later profile discrepancy described in Materials and Methods.

Family	Dice	Parameters (M)	Active TFLOPs
Canonical U-Net	0.902464	22.58	3.79–4.24
Attention-derived fixed	0.903489	22.21	3.57–4.01
Drop-derived fixed	0.902562	22.58	3.79–4.24
Multi-stage uniform	0.902626	26.04	8.56–9.00
Multi-stage SK	0.902913	26.07	8.56–9.00

Table 5. Recomputed paired patient-cluster sensitivity for static simplification. Intervals condition on stored models and preserve the original region-macro endpoint. No non-inferiority margin was specified.

Comparison	Difference	Conditional 95% interval
Attention-derived fixed − Multi-stage SK	+0.000576	[-0.000842, +0.002145]
Drop-derived fixed − Multi-stage SK	-0.000351	[-0.002279, +0.001669]
Attention-derived fixed − Canonical U-Net	+0.001025	[-0.000387, +0.002564]
Drop-derived fixed − Canonical U-Net	+0.000098	[-0.001515, +0.001822]
Multi-stage SK − Multi-stage uniform	+0.000287	[-0.001324, +0.001854]

Table 5 adds patient clustering to the original study-row bootstrap. Small differences and their intervals leave the preferred compact schedule unresolved. The separate five-fold single-configuration matrix also contains closely performing fixed/adaptive models; its terminal metrics provide architectural context in the manifest, not additional three-seed intervention replicas.

5.5. Selector information and functional reference

The supporting five-fold, 100-epoch, single-base-seed final-block dense-soft comparison is rebuilt in Table 6. Average-plus-maximum pooling increases study-level weight variation in all five folds. Equal-fold mean weight SD increases from 0.00159 to 0.04807, and the mean branchwise weight–removal correlation from 0.004 to 0.224; the correlation increases in four of five folds, not uniformly. Mean top-weight/forced-path winner agreement rises from 0.64% to 9.02%, but the average-plus-maximum values range from 0 to 23.5%: increased selector variation/alignment does not ensure correct forced-path ordering. Functional correlations and winner agreement use study-wise available-region Dice; model scores use region-macro Dice, with both reductions retained separately. Ties use the first maximum in fixed branch order. Historical GAP intervention baselines differ from terminal validation by at most 1.43×10^{-5}; both are retained, but effects use their own paired intervention baseline.

Table 6. Rebuilt supporting pooling comparison. Weight SD is the mean over branches of the across-study population standard deviation. The weight–drop column averages four branchwise Spearman correlations. Winner agreement compares largest weight and highest forced-path available-region Dice on each study.

Pooling	Fold	Dice	Weight SD	Weight–drop ρ	Winner agreement
GAP	0	0.900315	0.00188	-0.043	0.4%
GAP	1	0.907862	0.00155	0.101	0.4%
GAP	2	0.903934	0.00186	-0.067	0.8%
GAP	3	0.902920	0.00089	0.087	0.4%
GAP	4	0.898734	0.00178	-0.059	1.2%
Average + maximum	0	0.903226	0.05531	0.344	23.5%
Average + maximum	1	0.905870	0.05566	0.391	7.2%
Average + maximum	2	0.907032	0.03476	0.120	1.2%
Average + maximum	3	0.904282	0.04004	0.014	0.0%
Average + maximum	4	0.899314	0.05458	0.253	13.2%

5.6. Sensitivity to absent-region false positives

An all-case sensitivity scores every region of every study, assigning one to empty-reference/empty-prediction pairs and zero to empty-reference/nonempty pairs. Present-region Dice is unchanged. This makes absent-region false positives visible, while introducing a convention for correct empty predictions. Table 7 shows that weight–removal associations remain positive, while forced-path associations remain substantially weaker. The static-schedule contrasts in Appendix A likewise retain intervals spanning zero. These checks support the distinction between the functional targets under both scored endpoints; they do not equate Dice with clinical utility or quantify the volume of absent-region false positives.

Table 7. Stage-level endpoint sensitivity. Spearman correlations use the same twelve cells per fold as Table 3.

Fold	Positive-ref. drop	All-case drop	All-case keep
S1/0	0.951	0.944	0.021
S1/1	0.783	0.888	0.105
S1/2	0.944	0.944	-0.014
S1/3	0.909	0.797	0.021
S1/4	0.734	0.629	0.042
S2/0	0.252	0.385	-0.203
S2/1	0.552	0.566	-0.238
S2/2	0.790	0.741	-0.147
S2/3	0.587	0.643	-0.224
S2/4	0.804	0.538	0.014
S3/0	0.503	0.545	-0.021
S3/1	0.664	0.448	0.070
S3/2	0.804	0.853	0.273
S3/3	0.944	0.867	-0.189
S3/4	0.734	0.573	-0.154

6. Discussion

6.1. Interpreting a weight requires an operator and a target

Equation (10) separates the coefficient, branch-feature direction and amplitude, and downstream sensitivity. Normalization removes one class of variation while preserving centered-direction changes, allowing useful weight–removal association to coexist with poor weight–forced-path association.

A branch may be necessary in a co-adapted mixture yet weak alone, or strong alone yet redundant in the mixture. The counterexamples exclude universal coefficient ordering

of either target. Brain-tumor results demonstrate these distinctions under the implemented normalization/readout; rank adjustments and most stage exclusions retain the contrast while exposing local exceptions.

6.2. Finite epsilon makes removal-policy redundancy stage-dependent

Post-fusion instance normalization attenuates positive per-channel rescaling when surviving variance is large relative to epsilon and the edits preserve proportionality. Paired removals then provide nearly redundant block-level tests; counting them independently would overstate evidence. This prediction is conditional, not a checkpoint-wide equivalence.

The audit verifies both regimes: small later-stage differences and large stage-1 differences under epsilon-dominated surviving variance. Accurate finite-epsilon prediction, negligible zero-epsilon replay differences and persistent FP32 effects locate the block mechanism beyond paired-effect correlation. Check actual surviving variance and scale, not merely the presence of InstanceNorm: nearly proportional mixtures may diverge after normalization, and small block differences need not preserve thresholded outputs. The stage-level mechanism is not a full channel-to-mask decomposition.

Different normalization placement or spatial selection can preserve rescaling effects. Computational-graph checks should precede treating removal policies as separate functional tests.

6.3. Simplification needs a training comparison

Frozen keep-one inference tests an existing path, not its independently trained performance, consistent with the distinct targets of head ablation and removal/retraining [12,14]. Static schedules add compact-model training and terminal comparisons. Recorded compute savings remain conditional on the profiles; the similar canonical baseline and exploratory construction limit attribution to a diagnostic signal. Evaluate trained compact alternatives rather than infer architectural necessity from weights. Table 8 organizes the required evidence, not a validated optimal-schedule algorithm.

Table 8. Practical evaluation pathway for a learned kernel mixture. Choose the row matching the intended decision.

Intended decision	Evaluation to perform	Interpretation and next step
Identify dependence within the fitted mixture	Remove a branch; inspect post-fusion normalization and surviving scales.	Report the resulting metric change. Before treating renormalized removal as another distinct test, examine paired block and downstream responses.
Assess one path in the fitted network	Retain that branch at the selected stage, preserving the shared network.	Report forced-path performance. Independent training is needed to evaluate a compact architecture.
Choose a smaller trained architecture	Train compact candidates, including the canonical model and simple fixed schedules, with a matched recipe and explicit resource accounting.	Select on development data and evaluate on untouched patients. Compare the selected candidate with compact and dense baselines under the same endpoint.

Before final evaluation, specify acceptable performance loss, resource budget and uncertainty analysis. Select candidates meeting those criteria; otherwise revise candidates or retain the baseline. The approximately 53–58% recorded FLOPs reduction is against dense controls, requiring the similarly compact canonical comparator. This retrospective

study does not implement the proposed development/evaluation separation or establish prospective non-inferiority.

6.4. Relation to the broader expert-routing question

This paper addresses internal kernel signals under specified interventions and normalization. Stable expert roles, accessible targets and held-out adaptive-policy benefit require additional system-level evidence. Our claims are self-contained, without oracle-routing or external-model experiments.

7. Scope and Remaining Evidence

The mathematical identities rely on explicit operator conditions and use standard normalization algebra. Their role is to explain and test the intervention geometry, not to claim priority for scale invariance. The BraTS intervention study uses one cohort, all five folds, and three base-seed groups. The direction of the functional-reference distinction repeats across fifteen banks, but association strengths and some named branch rankings change between seeds. This bounds the reproducibility claim to the tested recipe and cohort. Averaging channel weights and studying stage-level cells can also hide channel-specific or case-specific effects.

The numerical operator controls use untrained tensors and an instrumented small selector; the completed observational audit additionally measures all fifteen trained checkpoints. Neither operation establishes how retraining with another normalization location would change segmentation performance. The FP32 sensitivity covers ten fixed patients rather than the full cohort; individual channels were not causally isolated at the final mask. The five-fold pooling results remain supporting single-seed evidence. The static schedules were chosen using the same folds later used for internal repetition, and no prospective non-inferiority margin was set. Conditional patient resampling addresses repeated studies, not architecture selection bias or optimization uncertainty. The all-case sensitivity makes empty-region errors visible but depends on the empty/empty scoring convention. Three seed groups do not characterize the full distribution of optimization variability; the five-fold static-schedule comparisons remain single-seed evidence and are not enlarged by the stage-study replication.

8. Conclusions

Kernel-selection coefficients describe an allocation inside a trained computational graph. Their functional meaning depends on the target of intervention and on the operators after fusion. Per-channel instance normalization makes paired removal differences small in a high-variance regime, but finite epsilon preserves substantial differences in low-variance first-stage mixtures. Direct checkpoint measurements and matched precision sensitivity verify this conditional interpretation. Removal dependence and forced-path performance remain distinct. All fifteen terminal brain-tumor banks across five folds and three base seeds exhibit stronger weight association with removal dependence than with forced-path performance, with seed-dependent association strength. In the separate single-base-seed discovery and internal-repetition comparison, compact static schedules achieve substantial recorded compute savings relative to dense multi-branch controls without identifying a uniquely favored schedule. Explicit operator analysis, paired interventions, and trained compact comparators provide a concrete basis for interpreting kernel weights. For practical use, distinguish a dependence audit, a frozen-path test, and independent compact-model training, then evaluate the selected structure against a compact baseline on patients not used for selection.

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Institutional Review Board Statement: This study is a secondary computational analysis of previously collected, de-identified BraTS 2023 research images and annotations. No participants were recruited or contacted, no clinical intervention was performed, and no personal identifiers were collected or recorded. The analysis meets the criteria for existing-data research eligible for exemption from institutional ethics review under Article 13(1)(3) of the Enforcement Rule of the Bioethics and Safety Act of the Republic of Korea (https://www.law.go.kr/lsLinkCommonInfo.do?chrClsCd=010202&lspttninfSeq=75929).

Informed Consent Statement: No participants were contacted and no new informed consent was collected for this secondary computational analysis. Access to the BraTS data is governed by the provider's data-use conditions; original data-collection consent is not represented as consent obtained for this study.

Data Availability Statement: A paper-specific reproducibility package containing code, versioned configurations, environment specifications, source-artifact hashes, study-level scores and selector weights, and lossless channel-level numerical measurements is provided as `mathematics-reproducibility-v1.0.0.zip` in the versioned release at https://github.com/jwon112/kernel-selection-normalization/releases/tag/v1.0.0. The maintained code repository is https://github.com/jwon112/kernel-selection-normalization. CPU scripts recalculate the reported bootstrap intervals, selector associations, normalization endpoints and channel distributions. BraTS images are not redistributed and require access under the provider's terms. Large trained checkpoints and full-resolution feature tensors are excluded; checkpoint provenance and regeneration code are supplied.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Static-Schedule All-Case Sensitivity

The same paired patient-cluster bootstrap is applied to the all-case endpoint defined in the Results. The region and study aggregation convention is fixed for all policies. This sensitivity uses the same fitted models and patients as the primary schedule analysis.

Table A1. All-case static schedule contrasts, including absent regions.

Comparison	All-case difference	Conditional 95% interval
Attention-derived fixed − Multi-stage SK	+0.000311	[-0.001596, +0.002245]
Drop-derived fixed − Multi-stage SK	-0.000339	[-0.002962, +0.002437]
Attention-derived fixed − Canonical U-Net	-0.000047	[-0.001862, +0.001755]
Drop-derived fixed − Canonical U-Net	-0.000698	[-0.003090, +0.001704]
Multi-stage SK − Multi-stage uniform	+0.000805	[-0.001506, +0.003058]

Appendix B. Training Augmentation Specification

The `nnunet_gpu` profile in `src/seg3d/data.py` applies the transformations in Table A2, in the listed order. Spatial transformations are shared across modalities and labels; intensity operations affect images only. Affine image interpolation is trilinear and label in-

terpolation is nearest-neighbor, with zero padding and `align_corners=False`. Contrast is centered on the spatial mean; gamma is applied after per-channel min–max normalization and mapped back to the original range. The blur is box averaging, not Gaussian smoothing.

Table A2. Strong full-volume training augmentation. Probabilities are Bernoulli application probabilities at the indicated sampling unit.

Operation	Probability/unit	Parameters
Affine	0.20 / study	Independent rotations on three axes, $[-30, 30]^\circ$; isotropic scale $[0.7, 1.4]$; no translation.
Axis flips	0.50 / study/axis	Independently flip each of three spatial axes.
Intensity	1 / study/channel	Scale $[0.75, 1.25]$ and shift $[-0.1, 0.1]$.
Noise	0.15 / batch	Add independent Gaussian noise, standard deviation 0.01.
Contrast	0.15 / study	Per-channel factor $[0.75, 1.25]$.
Gamma	0.15 / study	Per-channel exponent $[0.7, 1.5]$.
Box blur	0.20 / batch	3^3 average pooling, stride 1, padding 1.
Low-resolution simulation	0.25 / batch	Common factor $[0.5, 1.0]$; trilinear downsampling and restoration to the input shape.

Appendix C. Within-Stage Functional Measurements

Appendix C.1. Stage-adjusted and stage-exclusion sensitivities

Three retrospective within-bank checks assess dependence on stage mixing. First, rank weights and endpoints across twelve cells (average ranks for ties), subtract each stage's mean rank, then correlate the residuals by Pearson correlation. This residualization on categorical stage indicators adjusts ranks without intervening on network depth. Second, independently rank four branches per stage and correlate the concatenated vectors, removing between-stage rank magnitudes and offsets. Third, recompute ordinary Spearman correlations after omitting each stage, leaving eight cells.

Table A3 compares signed removal association with absolute forced-path association. The stage-adjusted contrast holds in 15/15 banks; independent within-stage ranks retain it in 14/15, except S2/fold 0 (removal -0.067, forced path -0.200). When stage 1 is omitted, S2/fold 0 ties (0.405 versus -0.405) and S2/fold 1 reverses the contrast (0.143 versus -0.286). Omitting stage 3 reverses it in S2/fold 0 (0.095 versus -0.143). Omitting stage 2 gives no reversal or tie. All stage-exclusion removal associations remain positive, a weaker requirement than exceeding the absolute forced-path association.

Table A3. Retrospective stage sensitivities on the fifteen terminal banks. Positive counts removal associations above zero. Stronger/tie/weaker counts compare signed removal with absolute forced-path association (tie tolerance 10^{-12}). Correlation columns report median [minimum, maximum] across banks. Each ordinary within-stage correlation uses four branches; adjusted and concatenated-rank correlations use twelve cells; exclusions use eight. Counts are descriptive, not independent binomial trials.

Check	Positive	Stronger/tie/weaker	Removal ρ	Forced-path ρ
All stages	15/15	15/0/0	0.783 [0.252, 0.951]	-0.021 [-0.238, 0.273]
Stage-adjusted ranks	15/15	15/0/0	0.815 [0.230, 0.966]	-0.012 [-0.239, 0.325]
Within-stage ranks	14/15	14/0/1	0.800 [-0.067, 1.000]	-0.133 [-0.333, 0.333]
Omit stage 1	15/15	13/1/1	0.810 [0.143, 0.976]	-0.167 [-0.524, 0.024]
Omit stage 2	15/15	15/0/0	0.762 [0.095, 0.952]	0.119 [-0.333, 0.524]
Omit stage 3	15/15	14/0/1	0.810 [0.095, 1.000]	-0.143 [-0.310, 0.357]
Stage 1 only	14/15	12/1/2	0.800 [-0.400, 1.000]	0.200 [-0.400, 1.000]
Stage 2 only	14/15	11/1/3	1.000 [-0.200, 1.000]	-0.400 [-0.800, 0.200]
Stage 3 only	13/15	10/1/4	0.800 [-0.600, 1.000]	-0.200 [-1.000, 0.800]

Renormalized removal retains the adjusted contrast in 15/15 banks and the independently ranked within-stage contrast in 14/15, with the same stage-exclusion counts as unnormalized removal. Under the all-case endpoint, both rank adjustments retain it in 15/15 banks; stage-exclusion counts are 15/15, 15/15, and 14/15. The all-case exception when stage 3 is omitted is S3/fold 1 (removal 0.238, forced path −0.310). Neither sensitivity replaces the primary endpoint. Generated evidence retains all bank-level values and exceptions; shared training data and validation patients preclude independent patient-evidence interpretations.

Appendix C.2. Individual stage correlations and functional ranges

Tables A4 and A5 disaggregate each fold's twelve cells from Table 3 into four-branch stage correlations and minimum–maximum ranges. These describe branch summaries, not patient ranges, confidence intervals or independent replicas, under the primary endpoint.

Table A4. Within-stage four-branch Spearman associations on fifteen terminal SK banks (S1–S3, five folds each). Decoder resolutions are given in Table 1.

Fold	Decoder	Drop ρ	Renormalized ρ	Forced-path ρ
S1/0	1	1.000	1.000	0.200
S1/0	2	1.000	1.000	-0.800
S1/0	3	0.800	0.800	0.200
S1/1	1	0.800	0.400	0.400
S1/1	2	0.800	0.800	-0.400
S1/1	3	1.000	1.000	0.400
S1/2	1	1.000	1.000	0.200
S1/2	2	1.000	1.000	-0.400
S1/2	3	1.000	1.000	0.400
S1/3	1	0.800	1.000	-0.200
S1/3	2	1.000	1.000	-0.400
S1/3	3	0.800	0.800	0.000
S1/4	1	0.800	0.800	0.400
S1/4	2	0.800	0.800	0.200
S1/4	3	0.800	0.800	-0.200
S2/0	1	-0.400	-0.400	0.200
S2/0	2	0.400	0.400	-0.400
S2/0	3	-0.200	-0.200	-0.400
S2/1	1	0.800	0.800	-0.400
S2/1	2	-0.200	-0.200	-0.400
S2/1	3	0.800	0.800	-0.200
S2/2	1	0.800	0.800	0.400
S2/2	2	0.400	0.400	-0.800
S2/2	3	0.800	0.800	-0.200
S2/3	1	0.400	0.200	0.000
S2/3	2	0.800	0.800	-0.400
S2/3	3	0.400	0.400	-0.600
S2/4	1	0.800	0.800	0.200
S2/4	2	1.000	1.000	0.200
S2/4	3	0.800	0.800	-0.200
S3/0	1	0.400	0.400	-0.400
S3/0	2	1.000	1.000	-0.400
S3/0	3	-0.600	-0.600	0.800
S3/1	1	1.000	1.000	-0.400
S3/1	2	0.200	0.200	-0.400

Fold	Decoder	Drop ρ	Renormalized ρ	Forced-path ρ
S3/1	3	0.600	0.600	0.800
S3/2	1	0.400	0.400	1.000
S3/2	2	1.000	1.000	0.200
S3/2	3	1.000	1.000	-0.200
S3/3	1	1.000	1.000	0.200
S3/3	2	1.000	1.000	0.200
S3/3	3	1.000	1.000	-1.000
S3/4	1	0.600	0.600	0.200
S3/4	2	1.000	1.000	-0.400
S3/4	3	0.800	0.800	-0.200

Table A5. Ranges across four branch summaries within each fold and stage. Effects are learned Dice minus intervened Dice; negative removal effects indicate an improvement. Forced-path values are Dice, not effects.

Fold	Decoder	Weight	Drop effect	Renorm. effect	Forced-path Dice
S1/0	1	[0.084, 0.441]	[0.000221, 0.0873]	[0.000144, 0.368]	[0.121, 0.879]
S1/0	2	[0.074, 0.398]	[0.000114, 0.104]	[0.000115, 0.104]	[0.137, 0.579]
S1/0	3	[0.090, 0.419]	[0.000698, 0.0133]	[0.000699, 0.0133]	[0.228, 0.898]
S1/1	1	[0.047, 0.552]	[-0.000299, 0.0106]	[-0.00017, 0.00262]	[0.747, 0.885]
S1/1	2	[0.047, 0.457]	[0.000285, 0.15]	[0.000285, 0.15]	[0.163, 0.895]
S1/1	3	[0.106, 0.405]	[0.000413, 0.00534]	[0.000412, 0.00533]	[0.611, 0.869]
S1/2	1	[0.018, 0.383]	[1.15e-05, 0.0386]	[1.69e-05, 0.0124]	[0.564, 0.898]
S1/2	2	[0.067, 0.417]	[0.00067, 0.116]	[0.00067, 0.116]	[0.214, 0.901]
S1/2	3	[0.098, 0.403]	[0.000188, 0.00561]	[0.000189, 0.0056]	[0.606, 0.898]
S1/3	1	[0.061, 0.445]	[0.00018, 0.0256]	[-7.27e-05, 0.0157]	[0.506, 0.803]
S1/3	2	[0.046, 0.447]	[-5.21e-05, 0.0973]	[-5.18e-05, 0.0973]	[0.155, 0.892]
S1/3	3	[0.109, 0.410]	[-0.000296, 0.00527]	[-0.000296, 0.00527]	[0.735, 0.895]
S1/4	1	[0.107, 0.475]	[-2.48e-05, 0.0341]	[-9.88e-05, 0.0348]	[0.714, 0.848]
S1/4	2	[0.072, 0.368]	[0.000437, 0.0736]	[0.000436, 0.0732]	[0.465, 0.889]
S1/4	3	[0.115, 0.399]	[0.00113, 0.00534]	[0.00113, 0.00534]	[0.815, 0.877]
S2/0	1	[0.099, 0.603]	[-0.000219, 0.000614]	[-0.000219, 0.000614]	[0.475, 0.902]
S2/0	2	[0.097, 0.394]	[0.00188, 0.0368]	[0.00188, 0.0368]	[0.326, 0.886]
S2/0	3	[0.141, 0.355]	[0.00398, 0.0122]	[0.00398, 0.0122]	[0.588, 0.860]
S2/1	1	[0.037, 0.446]	[1.64e-06, 0.434]	[-9.08e-05, 0.238]	[0.268, 0.898]
S2/1	2	[0.062, 0.401]	[-0.000106, 0.0231]	[-0.000146, 0.0231]	[0.514, 0.893]
S2/1	3	[0.119, 0.377]	[0.000379, 0.00273]	[0.000379, 0.00273]	[0.274, 0.874]
S2/2	1	[0.059, 0.547]	[4.53e-05, 0.0321]	[9.87e-05, 0.00473]	[0.668, 0.902]
S2/2	2	[0.046, 0.384]	[0.00025, 0.0116]	[0.000251, 0.0114]	[0.423, 0.893]
S2/2	3	[0.125, 0.393]	[0.000494, 0.00556]	[0.000494, 0.00556]	[0.605, 0.896]
S2/3	1	[0.116, 0.454]	[-0.000486, 0.00153]	[-0.000484, 0.000636]	[0.826, 0.861]
S2/3	2	[0.083, 0.468]	[9.92e-05, 0.12]	[9.95e-05, 0.12]	[0.109, 0.886]
S2/3	3	[0.131, 0.364]	[0.000589, 0.00637]	[0.000589, 0.00637]	[0.442, 0.857]
S2/4	1	[0.070, 0.399]	[-0.000713, 0.0826]	[-0.000534, 0.0859]	[0.189, 0.888]
S2/4	2	[0.047, 0.417]	[1.95e-05, 0.0492]	[1.97e-05, 0.0492]	[0.490, 0.886]
S2/4	3	[0.141, 0.390]	[0.000759, 0.00455]	[0.000759, 0.00455]	[0.478, 0.887]
S3/0	1	[0.073, 0.572]	[-0.000164, 0.0233]	[-0.000168, 0.0228]	[0.373, 0.903]
S3/0	2	[0.072, 0.476]	[0.000248, 0.22]	[0.000248, 0.22]	[0.047, 0.882]
S3/0	3	[0.081, 0.456]	[0.00576, 0.0333]	[0.00576, 0.0333]	[0.387, 0.840]
S3/1	1	[0.120, 0.568]	[-9.02e-05, 0.00408]	[-9.33e-05, 0.00411]	[0.557, 0.906]
S3/1	2	[0.101, 0.414]	[0.00101, 0.132]	[0.00101, 0.132]	[0.066, 0.875]
S3/1	3	[0.093, 0.424]	[0.000452, 0.0119]	[0.000452, 0.0119]	[0.718, 0.896]
S3/2	1	[0.092, 0.380]	[-0.000445, 0.113]	[0.000318, 0.0302]	[0.465, 0.892]
S3/2	2	[0.083, 0.396]	[0.000814, 0.248]	[0.000814, 0.248]	[0.145, 0.882]
S3/2	3	[0.148, 0.356]	[0.000929, 0.0036]	[0.000929, 0.00361]	[0.783, 0.895]
S3/3	1	[0.065, 0.465]	[-4.63e-05, 0.0309]	[-5.22e-05, 0.0393]	[0.496, 0.871]
S3/3	2	[0.056, 0.440]	[0.00018, 0.082]	[0.000179, 0.082]	[0.337, 0.881]

Fold	Decoder	Weight	Drop effect	Renorm. effect	Forced-path Dice
S3/3	3	[0.081, 0.412]	[0.000765, 0.00872]	[0.000765, 0.00871]	[0.543, 0.851]
S3/4	1	[0.107, 0.479]	[0.000213, 0.0027]	[0.000214, 0.00271]	[0.824, 0.886]
S3/4	2	[0.078, 0.405]	[-3.46e-05, 0.0532]	[-3.47e-05, 0.0532]	[0.445, 0.774]
S3/4	3	[0.083, 0.432]	[0.000326, 0.00533]	[0.000325, 0.00534]	[0.562, 0.845]

Appendix D. Evaluation Denominators and Run Correspondence

Table A6 lists eligibility shared by all five schedule models. Repeated-study patients contribute both eligible studies but one bootstrap multiplicity. A patient's region is present if either study has a positive reference, requiring separate study/patient denominators.

Table A6. Evaluation studies/patients and positive-reference region denominators, common to all five schedule-study models.

Fold	All studies/patients	WT studies/patients	TC studies/patients	ET studies/patients
0	251/227	251/227	250/227	246/223
1	250/226	250/226	248/225	243/220
2	250/226	250/226	250/226	245/221
3	250/227	250/227	248/226	243/221
4	250/227	250/227	249/227	241/219

Table A7 checks all twenty-five runs: batch size 4, base seed 2026, specified GPU augmentation, bfloat16, 100 epochs and terminal/latest evaluation. Actual and versioned configurations agree except machine-dependent dataset/output paths. This verifies recorded recipes, not unrecorded software/hardware identity.

Table A7. Recorded schedule-study run correspondence. Terminal epoch is one-based; the evaluation JSON records zero-based epoch 99. Differences compare each run with its own source configuration, preserving the intended architecture differences between families.

Family	Folds	Base seed	Terminal epoch	Config differences
Canonical U-Net	0–4	2026	100	None
Attention-derived fixed	0–4	2026	100	None
Drop-derived fixed	0–4	2026	100	None
Multi-stage uniform	0–4	2026	100	None
Multi-stage SK	0–4	2026	100	None

generated/run_reporting.json records model–fold run IDs, actual/source configuration paths and hashes, terminal epoch, checkpoint path and metric/profile CSV hash. Historical runs use results/brats2023_gli/<trainer>/fold_<f>/; added folds use results/mathematics_fold_completion_v1/brats2023_gli/<trainer>/fold_<f>/. The checkpoint is checkpoints/latest.pt; evaluation and profiling correspond to metrics/final_epoch_evaluation.json and metrics/fold_validation_final.csv, or metrics/fold_validation_final_epoch.csv for the added folds. Historical canonical folds 3–4 predate the evaluation JSON: checkpoint epoch/configuration and saved epoch-99 Dice are cross-checked directly. FLOPs retain each CSV's input shape/counting method. Added-fold and these two canonical checkpoints are content-hashed; other historical schedule checkpoints are identified by run/path, not newly content-hashed. The manifest explicitly maps canonical, attention-derived, drop-derived, uniform and SK trainer identifiers.

For five-fold/three-seed replication, generated/evidence.json hashes all fifteen terminal checkpoints and intervention artifacts. The build verifies completion-gate hashes, epoch 99 and fold study IDs, then reconstructs associations/winners from raw CSVs. Historical three-fold results are separate, not additional replication banks.

The S1 stage-replication and schedule-study SK banks have equal-fold means 0.904036 and 0.902913, respectively. In fold order 0–4, their Dice values are 0.903572/0.902341, 0.906557/0.904228, 0.907752/0.909127, 0.901784/0.899618 and 0.900514/0.899250 (replication/schedule). Replication uses KernelStageSK100Seed2026Trainer, under results/kernel_stage_fivefold_twoseed_v1/; schedules use BraTS23GLIMKMultiStageSK100Trainer, under the historical and missing-fold roots above. Configurations differ only in trainer, experiment and dataset/output paths. Frozen training/data/model/multi-kernel sources in the two recent campaigns are byte-identical; both pin the same current metadata/split hashes. Training enables cuDNN benchmarking without enforcing deterministic algorithms. Original runtime and processed-array byte identity for historical folds 0–2 remain unverified. The scores thus describe distinct fits, without isolating the difference's cause or proving historical execution equivalence.

Appendix E. Direct Normalization Diagnostics

The full-cohort exceedances in Table 2 are computed from sixty bank–branch cells per stage (three seeds, five folds, four branches), using the absolute difference of the two region-macro Dice endpoints. The six stage-1 cells above 0.01 comprise five dilated-branch removals and one anisotropic-branch removal. The three cells above 0.05 are dilated-branch removals in S1/fold 0, S2/fold 1 and S3/fold 2, with signed renormalized-minus-unnormalized differences of -0.280635, $+0.196149$ and $+0.083184$, respectively. Thus, the large differences are concentrated in one branch at one stage, repeat across all three seeds, and do not favor one removal policy uniformly. These thresholds are descriptive and were chosen after inspecting the audit, not prespecified hypothesis tests. The generated normalization evidence retains all endpoint cells and threshold counts stratified by seed, fold and branch.

Table A8 summarizes channel measurements at the edited stage. RMS is taken over spatial positions within one channel; maxima and minima are over the indicated channel records, not population intervals. The full-cohort audit contains 3,362,688 channel records; repeated patients, seeds and interventions are not independent samples. There are no zero surviving-mass records. The positive denominator floor is reached in 411 full-cohort stage-1 records and three stage-1 records in each precision subset; it is never reached at stages 2–3. Finite-epsilon predictions use the actual floored scale where applicable.

Table A8. Captured channel measurements across fifteen terminal banks. Post RMS is the observed difference between the two post-normalization outputs. Prediction RMS is the residual after applying the finite-epsilon identity to the captured unnormalized-removal output. Floored counts refer to positive surviving mass below 10^{-8}.

Mode/stage	Channels	Min v/ϵ	Max post RMS	Max prediction RMS	Floored
Full BF16/1	1,921,536	1.95e-13	1.00e+00	1.93e-06	411
Full BF16/2	960,768	1.88e+00	1.92e-01	1.95e-06	0
Full BF16/3	480,384	7.68e+01	6.15e-03	4.95e-06	0
Subset BF16/1	16,896	6.09e-12	1.00e+00	1.43e-06	3
Subset BF16/2	8,448	3.66e+00	1.14e-01	1.41e-06	0
Subset BF16/3	4,224	1.15e+02	4.31e-03	1.52e-06	0
Subset FP32/1	16,896	5.55e-12	1.00e+00	1.06e-06	3
Subset FP32/2	8,448	3.60e+00	1.15e-01	1.54e-06	0
Subset FP32/3	4,224	1.15e+02	4.30e-03	1.64e-06	0

Across full-cohort records, maximum pre-normalization relative scaling residuals are below 4.63×10^{-7}, and finite-epsilon prediction RMS residuals are below 4.95×10^{-6}. The corresponding maximum post-normalization RMS differences are approximately 1.000, 0.192 and 0.00615 at stages 1–3. In float64 replay, setting epsilon to zero gives a maximum paired RMS difference below 2.05×10^{-15} on the same captured features. These obser-

vations distinguish the finite-epsilon effect from a substantial failure of the proportional-mixture assumption.

For a descriptive case-level connection, S1/fold 0, study BraTS-GLI-01423-000, stage-1 dilated-branch removal changes available-region Dice by -0.759181 when renormalization is added. Its channel set has minimum variance/epsilon 4.23×10^{-10}, maximum post-normalization difference 0.999976 RMS, and maximum prediction residual 3.48×10^{-7}, with no denominator-floor event. This retrospectively selected largest case difference illustrates co-occurrence within one forward comparison, not independent confirmation or proof that a particular channel causes the full Dice change. The machine-readable generate d/normalization_evidence.json retains all 540 endpoint cells, stage/mode distributions, attempt identities and the largest paired case connections; raw per-case and per-channel files are bound by their completion SHA-256 hashes in generated/evidence.json.

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